

MDS-UPDRS Finger Tapping Analysis

The data provided below consists of extracted features (tap count, periods, and amplitudes) derived from a subject performing the MDS-UPDRS Item 3.4 Finger Tapping test. The subject was instructed to tap the index finger on the thumb 10 times as quickly and as largely as possible.

The objective is to score the subject according to MDS-UPDRS item 3.4: FINGER TAPPING based on these 10 consecutive finger taps. The testing instructions and scoring criteria for MDS-UPDRS item 3.4: FINGER TAPPING are as follows:

Instructions to the examiner:

Each hand is tested separately. Demonstrate the task, but do not continue to perform the task while the patient is being tested.

Instruct the patient to tap the index finger on the thumb 10 times as quickly AND as large as possible. Rate each side separately, evaluating speed, amplitude, hesitations, halts, and decrementing amplitude.

Scoring Criteria:

0 (Normal): No problems

1 (Slight): Any of the following:

- a) The regular rhythm is broken with one or two interruptions or hesitations of the tapping movement
- b) Slight slowing
- c) The amplitude decrements near the end of the 10 taps

2 (Mild): Any of the following:

- a) 3 to 5 interruptions during tapping
- b) Mild slowing
- c) The amplitude decrements midway through the 10-tap sequence

3 (Moderate): Any of the following:

- a) More than 5 interruptions during tapping or at least one longer arrest (freeze) in ongoing movement
- b) Moderate slowing
- c) The amplitude decrements starting after the 1st tap

4 (Severe): Cannot or can only barely perform the task due to slowing, interruptions, or decrements

Below are the extracted features including tap count, periods, and amplitudes. Please analyze this information and provide your score on the last line.

EXTRACTED FEATURES

Number of taps: 11

Tap periods (frames): [106, 158, 96, 134, 152, 129, 139, 112, 133, 153]

Tap amplitudes (pixels): [216.3631207022121, 213.92522057952868, 215.62931155109686, 219.49487465542333, 209.78083801910984, 203.8872237291979, 212.0, 217.13820483738002, 215.07673049402624, 210.27838690650069, 211.29126815843574]

GPT-5

MDS-UPDRS 3.4 Finger Tapping Scoring Analysis

Given data

- Number of taps detected: 11 (i.e., 10 intervals between taps)
- Tap periods (frames): [106, 158, 96, 134, 152, 129, 139, 112, 133, 153]
- Tap amplitudes (pixels): [216.36, 213.93, 215.63, 219.49, 209.78, 203.89, 212.00, 217.14, 215.08, 210.28, 211.29]

Note: 11 amplitude samples for 10 inter-tap intervals is consistent with 11 detected peaks.

1) Interruptions/hesitations

- Heuristics for hesitations: unusually long periods compared to the subject's own rhythm.
- Median period ≈ 136.5 frames; interquartile range is moderate.
- Longer intervals: 152 and 153 frames; there are 3–4 intervals ≥ 150 (158, 152, 153; plus 139/134 are near median). Clear outliers: 158; moderately long: 152, 153.
- Count of clear hesitations: approximately 3 (158, 152, 153). No evidence of any "longer arrest/freeze" (would expect a very large outlier).

Interpretation vs criteria:

- 3 to 5 interruptions → aligns with “2 (Mild)”.

2) Speed (slowing)

- Periods vary but stay within a band; there is no consistent progressive lengthening across the sequence that would signal clear slowing trend.
- Average speed appears mildly slowed only if the baseline is higher; with no normative frame-to-time mapping provided, we focus on within-sequence variability rather than absolute speed.

Interpretation:

- At most mild slowing (if any), consistent with score 1–2. Not clearly moderate.

3) Amplitude and decrement

- Amplitudes begin high (≈ 216 – 219) and dip around taps 5–6 (≈ 210 , 204), then recover to 217 and ~ 215 , later ~ 210 – 211 .
- Pattern: a mid-sequence dip with partial recovery, not a monotonic decrement from the first tap.
- Criteria mapping:
 - Decrement near the end → score 1
 - Decrement midway through the 10 taps → score 2
 - Decrement starting after the 1st tap → score 3
- The observed mid-sequence dip (around 5–6) with recovery best matches “decrements midway,” i.e., score 2.

Overall synthesis

- Interruptions: approximately 3 → score 2
- Speed: at most mild slowing → score 1–2
- Amplitude: decrement midway with partial recovery → score 2

Final clinical judgment: The combination of 3 hesitations and a clear mid-sequence amplitude decrement supports “2 (Mild)” rather than “1 (Slight)” or “3 (Moderate).”

Score: 2
